Detection and Localization of Closely Spaced Pipelines Using a 3-D Multistatic Subsurface SAR

Abdulrahman Aljurbua, *Member, IEEE*, and Kamal Sarabandi, *Life Fellow, IEEE*

***Abstract*— Detection and localization of buried pipelines is an important problem for oil and water distribution industries as well as for excavation and construction companies. A novel 3-D forward scattering model and dynamic grid search-based inversion algorithm of multiple buried pipelines using multistatic synthetic aperture radar (SAR) configuration is demonstrated in this article. Traditional imaging algorithms that focus the SAR signal by compensating only for the phase work well when the targets are far away from each other. By considering both the phase and amplitude of the scattered field and casting the inversion problem as an optimization problem, the detection and localization resolutions are improved for extended targets. A forward scattering model for multiple adjacent pipes is developed to track the scattering phase centers (SPCs) of each pipeline as the receiver location is changed. The model is used to calculate the resulting propagation phase and amplitude attenuation due to propagation in lossy media. The misfit between the measured data and the forward model (FM) is used as the cost function. The cost function is minimized by the dynamic grid search algorithm that starts with an initial focus and then performs a more careful search to detect closely spaced pipelines. The FM and inversion algorithm are validated and compared with a basic focusing algorithm using full-wave simulations as well as experimental data for both metallic as well as dielectric pipes. The results show improvement in range and cross-range detection and localization resolutions over the basic SAR focusing algorithm.**

***Index Terms*— Inverse problems, multiple pipeline detection, multiple pipeline localization, subsurface synthetic aperture radar (SAR).**

I. Introduction

MAPPING underground infrastructure is important for many applications ranging from archeology to transportation of various liquids using pipelines and various construction industries. Due to their low attenuation, long-wavelength electromagnetic (EM) waves can penetrate through soil and sense buried objects. The common technology employing EM waves for the detection of buried objects is ground-penetrating radars (GPRs) which has a vast treatment in the literature [1], [2], [3], [4], [5], [6]. For a typical GPR system, a transceiver attached to an antenna is moved above the search area and the signal is recorded as a function of delay for different positions. When the GPR crosses a buried pipeline, a hyperbola is generated in the delay-position radar response. Analyzing such hyperbola leads to the detection and localization of the buried pipeline. In this hyperbola-based detection approach, some research [7], [8], [9], [10] is overtaken to develop an accurate scattering model to predict the geometrical properties of the hyperbola resulting from a single pipeline to help to interpret the radar response in realistic scenarios. On the other hand, other researchers [11], [12], [13], [14] employ pattern recognition and machine learning techniques on the radar response to obtain a robust and automated hyperbola recognition and hence buried pipeline detection. The main issue with hyperbola recognition-based detection is its 2-D nature. This means that approach assumes a monostatic radar configuration or a special multistatic radar configuration where the receiver is confined along a straight line. This is not a major drawback when the response of the pipeline is strong enough leading to detection. However, when the pipeline response is rather weak, such as in the situations of small-diameter pipelines or deeply buried pipelines, detection may not be straightforward. To increase the probability of detection in these situations, a multistatic synthetic aperture radar (SAR) configuration with arbitrary receiver locations can be used to collect the pipeline's response from multiple angles. After that, the collected responses can be processed together coherently to obtain high processing gain that leads to improved detection. Particularly, the multistatic subsurface SAR system proposed in [15] and illustrated in Fig. 1 might be used. This system processes the collected signals coherently using the well-known back-projection (BP) focusing algorithm to form an image of the underground. However, BP works well

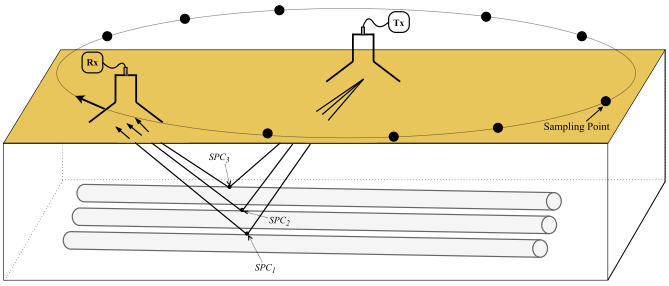


Fig. 1. Multistatic subsurface SAR system proposed in [15]. There is a fixed Tx illuminating the underground and a moving Rx that samples the scattered field around the Tx. It is noted that, unlike point targets that have a fixed SPC, pipelines have dynamic SPCs that depend on the Rx position relative to the Tx.

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only for point targets and to make it suitable for extended objects such as pipelines, a special focusing algorithm such as the one proposed in an earlier work [16] might be used to track the scattering phase centers (SPCs) of pipelines and compensate for the resulting phase appropriately to enable robust detection. Such an algorithm works fine for the detection of a single pipeline or widely spaced pipelines. However, as all imaging algorithms, this method also has finite resolution along the range and cross-range. Hence, the detection and identification of closely spaced pipelines require additional investigation. To get a better resolution, an optimization-based inverse scattering approach must be considered. Inversion-based approaches for better target detection are prevalent in the literature [17], [18], [19], [20], [21], [22], [23], [24], [25]. In these approaches, the cost function to be minimized is defined as the misfit between the measured radar response and the predicted response based on some forward scattering model. In this article, a novel 3-D forward scattering model for multiple buried pipelines in the multistatic subsurface SAR configuration is proposed. The objective function is minimized by a dynamic grid search-based algorithm to enable the detection and localization of multiple closely spaced pipelines. Due to the utilization of both the amplitude and the phase of the scatterers, this objective function approach outperforms an approach we developed earlier [16]—that searches and compensates only for the phase of the bistatic scattered field—by providing a better spatial resolution. The rest of the article is organized as follows. Section II starts with developing the forward scattering model for multiple buried pipelines and moves to explain the dynamic grid search-based inversion algorithm. Section III provides numerical analysis for some of the approximations made in the forward model (FM) derivation, and Section IV provides a comprehensive numerical and experimental evaluation of the novel dynamic grid search algorithm performance.

II. Multiple Buried Pipelines Detection and Localization Algorithm

The geometry of the problem under consideration is shown in Fig. 1. This figure shows multiple pipelines buried in a half-space dielectric medium with the pipes' axes parallel to the half-space interface. The transmitter (Tx) and receiver (Rx) antennas are assumed to be very close and matched to the half-space so that the antenna can be assumed to radiate spherical waves inside a single medium, which can be locally approximated as plane waves when considering the phase propagation behavior along a specific direction. The forward scattering model and inversion algorithm for a single pipeline in subsurface multistatic SAR configuration has been discussed in detail in the earlier work [16], they will be discussed briefly in Sections II-A–II-C to facilitate the derivation of the scattering model and inversion algorithm for the general case of multiple buried pipelines for the multistatic subsurface SAR.

A. Forward Scattering of a Single Pipeline

For $e^{j\omega t}$ time dependence and plane wave incidence at an angle θ_i with respect to the pipeline axis (z-axis), the analytical

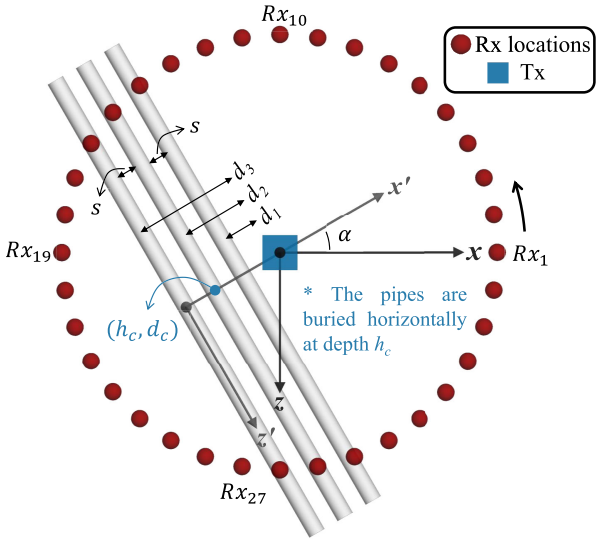


Fig. 2. Top view of the coordinate systems (the global and the primed one along one of the pipes) and the subsurface SAR configuration. The pipes have the same depth h_c with varying displacements determined by the spacing s. For vertically positioned buried pipes, the displacement will be d_c and the depths will be varying. The displacement direction is from the Tx location to a specific pipes' center.

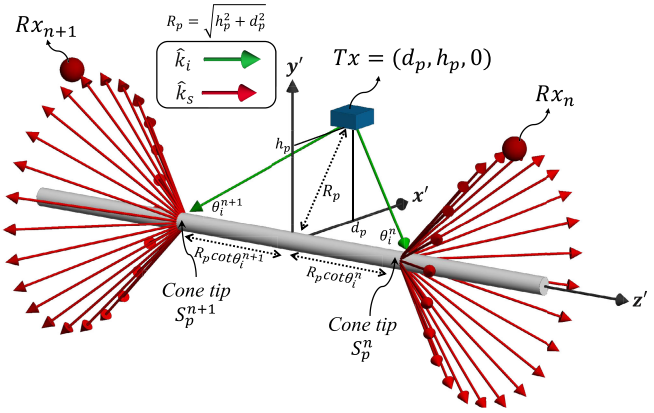


Fig. 3. Primed local coordinates along an arbitrary pipe p of depth h and displacement d from the Tx. Only two cones are shown for illustration. Each Rx has a unique cone and SPC (the cone tip). The same applies to the other pipes.

scattered field of a pipeline in the far field can be written as

$$\boldsymbol{E}^s \approx \boldsymbol{A}(\rho, \phi, \theta_i)\, e^{-j(k\cos\theta_i z + k\sin\theta_i \rho)} \tag{1}$$

where k is the wavenumber for the background medium. The vector $\boldsymbol{A}$ specifies the components of the scattered field that satisfy the boundary conditions for TE^z and TM^z modes and can be found analytically for various types of pipes [26], [27], [28], [29], [30]. The exponential term in (1) indicates that the scattered field will propagate along a surface of a cone until reaching a specific Rx location. The Tx antenna typically has a wide beam resulting in the generation of multiple scattered field cones (i.e., a cone for each possible wave incidence at θ_i as illustrated in Figs. 2 and 3). The cone tip for a wave coming at an angle θ_i is the location of the SPC for the particular Rx location that happens to intersect with the θ_i cone. To find the particular θ_i^n for the nth Rx location (hence the cone tip

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and the SPC), suppose the pipe's orientation angle is α, and its displacement and depth with respect to the Tx are d and h, respectively (see Fig. 2). In the primed coordinate system whose z-axis is along the pipeline (Fig. 3), the solution for θ_i^n is found by [16]

$$\theta_i^n = \arctan\left(\frac{\sqrt{h^2 + x_{\text{on}}'^2(\alpha, d, h)} + \sqrt{d^2 + h^2}}{z_{\text{on}}'(\alpha, d, h)}\right) \tag{2}$$

where $(z_{\text{on}}', x_{\text{on}}')$ are the coordinates of the Rx location in the pipeline coordinate system. These coordinates are functions of (α, d, h) parameters. Having found θ_i^n, the SPC coordinate is given by $S^n = (0, 0, \sqrt{d^2 + h^2}\cot\theta_i)$. From S^n, the total bistatic signal path length which consists of two segments can be calculated. The first segment is $R_{\text{Tx}-S^n}$, the distance from the Tx to the SPC, and the second segment is $R_{S^n-\text{Rx}_n}$, and the distance from the SPC to the nth Rx location. Knowing the path length allows for predicting the phase behavior which leads to the detection of the pipeline will be shown in Section II-C.

B. Forward Scattering of Multiple Pipelines

To model the scattering behavior of multiple pipelines, both the amplitude and phase of the scattered field from the pipelines need to be accounted for. To do that, the knowledge of the SPC of each pipeline for each Rx location is utilized to form the FM. For a collection of Np pipelines oriented at α, positioned horizontally or vertically (ar $= 1, 2$) with their centroid at (h_c, d_c) and their spacing being s (see Fig. 2), the FM of the received voltage at mth frequency and nth Rx location is given by

$$\text{FM}_{mn}(\text{ar}, N_p, d_c, h_c, s) = \sum_{p=1}^{N_p} P_{\text{Tx}}^m \times P_{\text{Rx}_n}^m \times \frac{e^{-\text{j}k_m(R_{\text{Tx}-S_p^n} + R_{S_p^n-\text{Rx}_n})}}{R_{\text{Tx}-S_p^n} \times R_{S_p^n-\text{Rx}_n}}. \tag{3}$$

Here, P_{Tx} refers to the Tx antenna radiation pattern that implicitly depends on $(\phi_p^{\text{Tx}}, \theta_p^{\text{Tx}})$ the coordinates of the SPC of the pth pipe as seen by the Tx spherical coordinate system centered at the Tx antenna centroid. The same applies to P_{Rx_n}, the Rx antenna radiation pattern that depends implicitly on $(\phi_p^{\text{Rx}_n}, \theta_p^{\text{Rx}_n})$ the coordinates of the pth pipe's SPC as seen by the spherical coordinates system centered at the Rx antenna centroid at the nth Rx location. Furthermore, the path in the exponent $(R_{\text{Tx}-S_p^n} + R_{S_p^n-\text{Rx}_n})$ accounts for the propagation phase and attenuation due to loss in the medium while the product $R_{\text{Tx}-S_p^n} \times R_{S_p^n-\text{Rx}_n}$ in the denominator accounts for the attenuation due to the spherical propagation of the waves emitted by the Tx. Depending on the arrangement ar and the spacing s, the values of depth h_p and displacement d_p of the pth pipeline can be obtained from

$$d_p = \begin{cases} d_c + \left(p - \dfrac{N_p + 1}{2}\right) \times s, & \text{ar} = 1 \\ d_c, & \text{ar} = 2 \end{cases} \tag{4}$$

$$h_p = \begin{cases} h_c + \left(p - \dfrac{N_p + 1}{2}\right) \times s, & \text{ar} = 2 \\ h_c, & \text{ar} = 1 \end{cases} \tag{5}$$

Algorithm I Basic focusing based pipelines detection

Input: $\{V_{11}, \ldots, V_{MN}\}, \{(x_{o1}, z_{o1}), \ldots, (x_{oN}, z_{oN})\}$, $\{(\alpha_1, d_1, h_1), \ldots, (\alpha_K, d_K, h_K)\}$,

For *k=1 to K* **Do**

- Transform (x_{on}, z_{on}) into (α_k, d_k, h_k) coordinates $\forall n$
- Solve for θ_i^n $\forall n$
- Calculate R_{Tx-S^n} and $R_{S^n-Rx_n}$ $\forall n$
- Compensate for the pipeline phase and do a coherent sum
- Save the sum in $Search(\alpha_k, d_k, h_k)$

End

$$(\alpha, d, h)_{Actual} = \underset{(\alpha,d,h)}{\text{argmax}} |Search(\alpha, d, h)|$$

Fig. 4. Basic phase compensation (focusing) algorithm for finding single pipeline or widely spaced pipelines parameters proposed in [16].

where (d_c, h_c) is the centroid of the collection of pipes. This FM is to be used to minimize an objective function leading to the detection and localization of the multiple buried pipeline as will be discussed in the following Section II-C.

C. Inversion Algorithms

Given the received voltage at each Rx location and frequency $\{V_{mn}\}$ in addition to the precise knowledge of the Tx and Rx locations, the aim of the inversion algorithms is to determine the parameters (α, h, d) (i.e., orientation, depth, and displacement) of each buried pipeline. One way to go about finding the parameters is to define the following basic focusing search:

$$\text{Search}(\alpha, d, h) = \sum_{m=1}^{M}\sum_{n=1}^{N} V_{mn} e^{j\left(k_m\left(R_{\text{Tx}-S^n} + R_{S^n-\text{Rx}_n}\right)\right)}. \tag{6}$$

This search simply compensates for the phase of a pipeline with parameters (α, h, d) across all Rx locations and frequencies and then sums the results coherently. The process can be repeated for every possible parameter (α, h, d), and the sum will be maximized at the actual pipeline parameters as illustrated in the algorithm in Fig. 4. This approach works fine if there is only a single buried pipeline. However, when there are multiple pipelines, especially closely spaced ones, the algorithm will no longer be able to separate them. This is because the algorithm is a focusing-based algorithm that entails a limited range and cross-range resolutions (full-width at half maximum of $|\text{Search}(\alpha, d, h)|$). Moreover, due to its relatively stronger scattering, the pipeline closest to the Tx will have a strong peak that may overshadow the peaks associated with the other pipelines. To overcome this limitation, the following objective function can be defined:

$$J\left(\text{ar}, N_p, d_c, h_c, s\right) = \sum_{m=1}^{M}\sum_{n=1}^{N} |V_{mn} - a_m \text{FM}_{mn}| \tag{7}$$

where a_m is an amplitude normalization factor calculated according to the following formula:

$$a_m = \frac{\max_n(|V_{mn}|)}{\max_n(|\text{FM}_{mn}|)}. \tag{8}$$

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Algorithm II Dynamic grid search based pipelines detection

Input: $\{V_{11}, \ldots, V_{MN}\}, \{(x_{o1}, z_{o1}), \ldots, (x_{oN}, z_{oN})\}, \max(N_p), \max(s), \delta s, \delta d, \delta h$

Do a search using **Algorithm I** to get the pipelines orientation α and an initial estimate of (h_c, d_c) i.e (h_c^0, d_c^0)

For $ar = [1,2], N_p = [1, \max(N_p)]$ **Do**
  For $s = [0, \max(s)]$ **with** δs spacing **Do**
    If ar=1
      ➢ $h = s$, $d = \frac{N_p}{2} s$
    Else
      ➢ $d = s$, $h = \frac{N_p}{2} s$
    End
    For $h_c = [-h, h] + h_c^0$ **with** δh spacing **Do**
      For $d_c = [-d, d] + d_c^0$ **with** δd spacing **Do**
        ➢ Calculate and store $J(ar, N_p, h_c, d_c, s)$
      End
    End
  End
End
$(ar, N_p, h_c, d_c, s) = \underset{(ar, N_p, h_c, d_c, s)}{\operatorname{argmin}} J(ar, N_p, h_c, d_c, s)$

Fig. 5. Novel dynamic grid search algorithm for finding multiple closely spaced pipeline parameters.

The objective function represents the misfit between the measured voltage data $\{V_{mn}\}$ and the predicted scattering behavior for a collection of pipelines with parameters (ar, N_p, d_c, h_c, s). When the pipeline parameters match the actual ones, the objective function is expected to be minimized leading to the detection and localization of multiple closely spaced pipelines. This objective function has five parameters, to minimize it in a computationally efficient manner, a dynamic grid search algorithm is applied to it. The dynamic grid search algorithm loops over all the discrete values of the parameters ar and N_p as well as the spacing s that has a limited range. To search for the pipelines collection centroid (d_c, h_c), the algorithm in Fig. 4 is applied first and the resulting peak in |Search(α, d, h)| is found to determine the pipelines orientation α as well as an initial estimate of (d_c, h_c). This is because, for closely spaced pipelines that are not resolved by the basic search, a single peak will appear within the spatial extent of the pipeline collection. After obtaining an initial estimate, a dynamic search window is applied based on the pipes number N_p and their arrangement ar. This means for horizontally (vertically) positioned buried pipes, the window will extend around d_c (h_c) dimension, where the extent is determined based on the number of assumed pipes N_p and the spacing s. The algorithm steps are summarized in Fig. 5.

III. Numerical Analysis

In the derivation of the FM, several approximations have been made to reach the final result. One approximation is that the pipelines are in the far field of the Tx and Rx antennas and vice versa. Another approximation is ignoring the scattering phase contribution from the vector $\boldsymbol{A}$ in (1) and using only the propagation phase term. These two approximations have been discussed in detail in the earlier work [16] and will not be discussed here. Aside from these approximations, the FM assumes the antennas to be very close to the soil and matched to it allowing for the approximation that the pipelines are buried in a uniform medium instead of the real scenario of an air medium above a soil half-space medium. In addition, the FM is a first-order scattering model that ignores the multiple scattering among pipelines and any possible shadowing effects. This section discusses the impact of the uniform medium assumption and the discardation of higher order scattering mechanisms. The numerical analysis and simulations throughout this article use one of three representative soil media summarized in Table I where the water content is obtained from the survey in [31] and the dielectric constants are calculated using Code 4.7 provided in [32].

TABLE I
Soil Media Used in Simulations and Numerical Analysis

Soil Type	Composition (Sand, Slit, Clay)	Relative Permittivity	Water Content
Sand	(96%,2%,2%)	$4.35 - j0.3$	1%
Loam	(12%,44%,44%)	$6.1 - j2.1$	10%
Clay	(2%,41%,57%)	$15.5 - j6$	30%

A. Uniform Medium Assumption

The reason for neglecting the half-space interface and treating the Tx and Rx antennas as if they were radiating and receiving through a uniform medium is based on the fact that when the antennas are placed very close to the soil interface and matched to it, the propagation phase goes mostly through the soil and only the amplitude radiation patterns of the antennas are effected by the interface presence. To demonstrate the validity of this assumption, the radiation patterns of the antennas can be obtained by simulating the antennas above a soil medium using a full-wave simulation. In this way, the effect of the air–soil interface becomes included in the radiation patterns in (3). To test this approach, Altair FEKO simulation software is used to generate full-wave scattered field data of two perfect electric conductor (PEC) pipelines of 10 cm diameter buried in a half-space dielectric medium. The Tx–Rx configuration is the same as the circular one shown in Fig. 2 with the circle having a 3.2 m diameter. The two pipes are at 1 m depth and oriented along the z-axis and placed at 0.5 and 1 m along the x-axis (i.e., $d_1 = -50$ cm, $d_2 = -100$ cm, $\alpha = 0°$, $h_c = -100$ cm). The antennas used for Tx and Rx are Hertzian dipoles oriented along the z-axis and placed 5 cm over the half-space consisting of air and sand medium in Table I. The FM is then used to find the scattered field to be compared with the full-wave solution. The FM is used twice. One time, the radiation patterns of the dipoles are obtained with analytical expressions for a dipole in a uniform medium [33] and the other time full-wave simulation is used to find the radiation patterns of the dipoles when placed above the half-space. The comparison results for the received voltage values for different Rx locations at 300-MHz frequency

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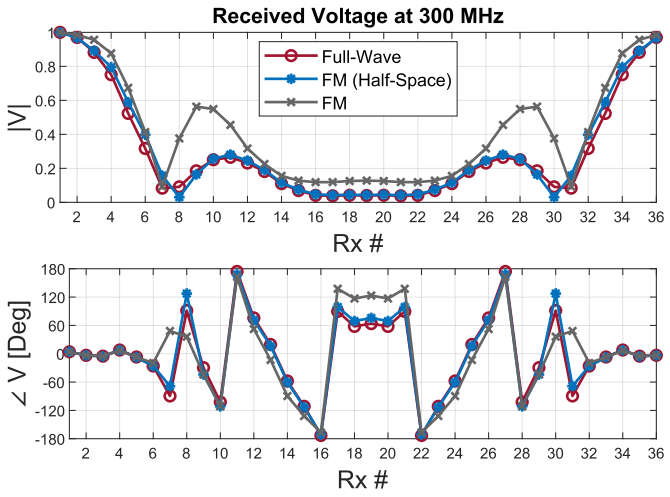


Fig. 6. Received voltage values for different Rx locations obtained by full-wave simulation as well as using the FM in (3) where the radiation patterns are calculated one time for a single medium—FM—and another time for half-space scenario—FM (half-space). The data are for two 10 cm diameter PEC pipelines with $d_1 = -50$ cm, $d_2 = -100$ cm, $\alpha = 0°$, and $h_c = -100$ cm (see Fig. 2).

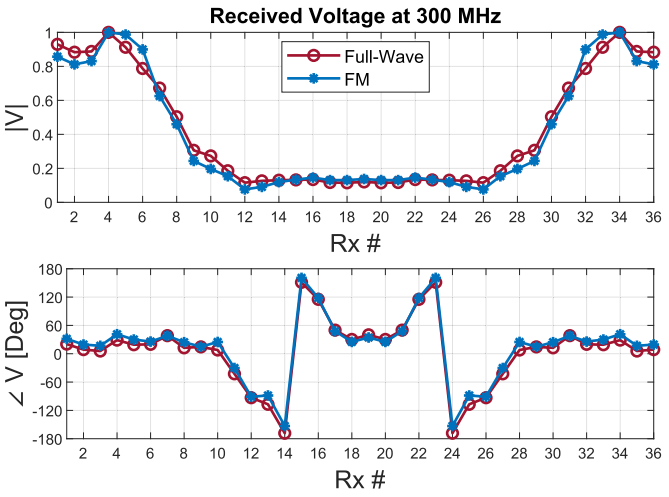


Fig. 7. Received voltage values for different Rx locations obtained by full-wave simulation as well as using the FM (3) at 300-MHz frequency. The data are for four horizontally positioned PEC pipelines of 29.1 cm diameter and their parameters are $d_1 = -13$ cm, $d_2 = -44$ cm, $d_3 = -75$ cm, $d_4 = -106$ cm, $\alpha = 0°$, and $h_c = -80$ cm (see Fig. 2) inside Loam medium.

(the central frequency of the radar used in this work) are shown in Fig. 6. The comparison shows that when calculating the radiation patterns assuming a uniform medium, noticeable deviations in amplitude from the full-wave data are observed. However, when using the radiation patterns of a dipole above the half-space the amplitude deviations are no longer observed and excellent agreement is obtained between the FM and the full-wave data. This indicates the validity of approximating the half-space with a single medium as long as the antennas are closed to the half-space interface and their amplitude radiation patterns are obtained when they are placed above the half-space medium.

B. Efficacy of the First-Order Scattering Assumption

To examine the efficacy of a first-order scattering model that ignores the mutual interactions among the pipelines, Altair FEKO simulation software is used to generate the full-wave scattered field data for two scenarios of the array of pipelines that are buried horizontally and vertically. The medium of choice for both scenarios is the moderately lossy Loam medium in Table I while the Tx–Rx configuration is the same as the one in Fig. 2 with 3.2 m diameter circle with the Tx and Rx antennas being Hertzian dipoles directed along the z-axis. For the horizontally positioned pipelines scenario, four standard size 8.625 in (21.9 cm) diameter [34] PEC pipelines with 10 cm edge-to-edge spacing are chosen as a representative of an example of relatively large pipes in terms of wavelength inside the Loam medium (about $(\lambda/2)$ diameter). The pipelines are buried at 80 cm depth below the Tx and oriented along the z-axis (i.e., $d_1 = -13$ cm, $d_2 = -44$ cm, $d_3 = -75$ cm, $d_4 = -106$ cm, and $\alpha = 0, h_c = -80$ cm in Fig. 2). The comparison results between the full-wave simulation and the FM (3) are shown in Fig. 7. It is observed that the amplitude value is rather low for Rx locations 10–27 which is expected as these Rx locations are the farthest from the pipelines making their scattered signals travel longer distances and hence experience more attenuation from the lossy Loam medium. Aside from that, excellent agreement with full-wave data is observed with slight expected deviations due to ignoring the mutual interactions among the pipelines. It will be shown in the performance evaluation section (Section IV) that such slight deviations will not prevent the cost function to be minimized at the correct parameters values leading to successful detection and localization. The other scenario is for three vertically positioned dielectric pipelines with inner and outer layers. The inner layer material is water with a dielectric constant of $80 - j20$ and standard diameter size of 4 in (10.2 cm) [34] (about $(\lambda/4)$ inside Loam). The outer layer represents polyvinyl chloride (PVC) material with a dielectric constant of 3 and a diameter of 10.204 cm. The three pipes are placed 1.25 m at the x-axis and oriented along the z-axis with 15 cm edge-to-edge vertical spacing (i.e., $h_1 = -75$ cm, $h_2 = -100$ cm, $h_3 = -125$ cm, $\alpha = 0°$, and $d_c = -125$ cm in Fig. 2). The comparison between the full-wave simulation data and the FM (3) at 300 MHz is shown in Fig. 8. Here as in the PEC pipes case, the low amplitude is observed for Rx locations 10–27. Moreover, excellent agreement is obtained in comparison with full-wave data with slight overestimation in amplitude especially at Rx locations 1–6 and 31–36 which are the locations closest to the pipelines as can be seen in Fig. 2. This slight overestimation of amplitude by the FM can mainly be attributed to excluding the effects of shadowing. These deviations should not prevent the detection and localization of these vertically buried pipelines as will be shown in Section IV.

IV. Performance Evaluation for the Inversion Algorithms

With the FM validated in Section III, this section aims to provide a comprehensive analysis of the dynamic grid search inversion algorithm performance. At first, the algorithm is

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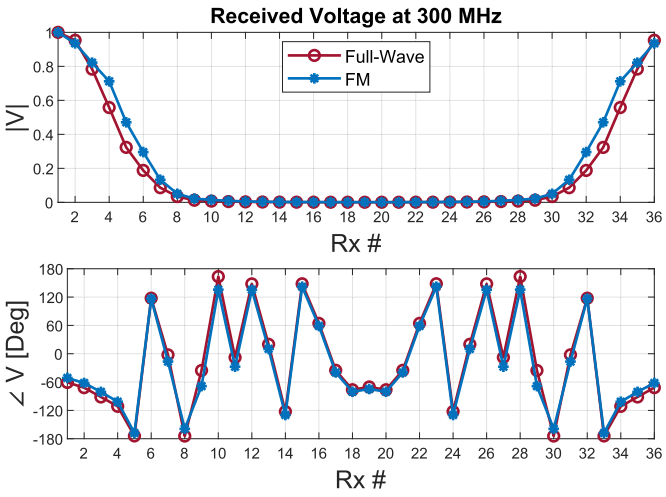
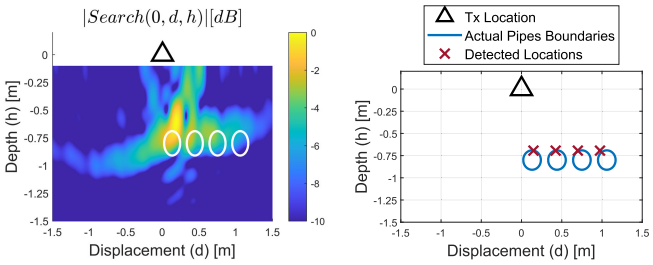


Fig. 8. Received voltage values for different Rx locations obtained by full-wave simulation as well as using the FM (3) at 300-MHz frequency. The data are for three dielectric pipelines of 10.2 cm diameter positioned vertically and their parameters are $h_1 = -75$ cm, $h_2 = -100$ cm, $h_3 = -125$ cm, $\alpha = 0°$, and $d_c = -125$ cm (see Fig. 2) inside Loam medium.

tested on results based on the full-wave simulation scenarios considered in the previous section. After that, the inversion error as a function of pipeline separation is investigated. Moreover, the robustness of the algorithm is tested and discussed by including the leakage signal from the Tx to the Rx as well as additive noise. Finally, the algorithm is evaluated on experimental data for metallic and water-filled PVC pipes buried in a realistic lossy sand environment.

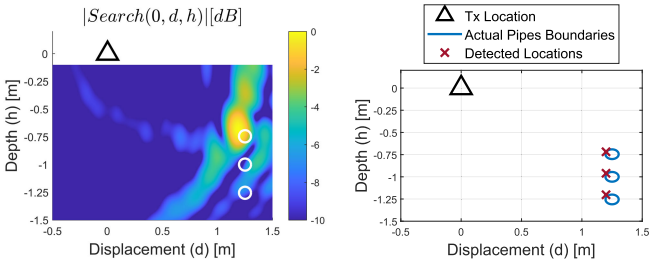
A. Performance Evaluation Using Full-Wave Simulation

To evaluate the dynamic grid search inversion algorithm performance, the two scenarios of four horizontally positioned PEC pipelines and three vertically positioned water-filled PVC pipelines in Loam medium in Section III-B are considered. The Tx and Rx arrangement is shown in Fig. 2 with an Rx circle diameter of 3.2 m. The Tx and Rx antennas are the same Hertzian dipoles and the simulation frequency range is from 150 to 450 MHz with a 20-MHz frequency step. Only the scattered field data are calculated with FEKO simulation software. The data are then processed using Algorithm I (Fig. 4) and Algorithm II (Fig. 5). The algorithms performance results for the first scenario of four horizontally positioned PEC pipes are shown in Fig. 9. The search results of applying Algorithm I show a clear peak near the boundary of the pipeline closest to the Tx location. However, the other pipelines are not detected as their responses are suppressed by the response from the pipe closest to the Tx. Moreover, for the third and fourth pipes, only one small peak is observed near them due to the resolution limitations of Algorithm I. When applying Algorithm II, the final detection results show the ability of the algorithm to detect all four pipelines where the detected locations correspond to the upper edges of the pipes indicating the physical fact that the first-order scattering occurs at the pipe's edges, not their centers. The results for the second scenario are shown in Fig. 10. Here also similar to the first scenario, only one peak is shown near the top pipeline



(a) (b)

Fig. 9. (a) h-d cut of the search space of Algorithm I (Fig. 4). (b) Detection results of Algorithm II (Fig. 5) when applied to full-wave simulation data. The data are for four PEC pipelines of 21.9 cm diameter and 10 cm edge-to-edge spacing inside Loam medium from Table I.



(a) (b)

Fig. 10. (a) h-d cut of the search space of Algorithm I (Fig. 4). (b) Detection results of Algorithm II (Fig. 5) when applied to full-wave simulation data. The data are for three water-filled PVC pipelines of 10.2 cm diameter and 15 cm edge-to-edge spacing inside Loam medium from Table I.

whereas the other pipelines are not detected by the basic search of Algorithm I. However, the results of applying Algorithm II show the ability of the algorithm to detect all three pipes at their edges closest to the Tx location.

B. Performance Evaluation With Pipelines Separation

In the derivation of the inversion algorithm (Algorithm II), the multiple scattering phenomena between the pipelines are ignored and only the first-order scattering was taken into account. This is because the pipelines are typically buried in a lossy soil medium that attenuates the second-order signals further because they travel longer paths compared with the first-order signals. To investigate this assumption, several full-wave numerical simulations are performed for horizontally and vertically positioned pipelines with varying separation using FEKO simulation software. For the horizontally positioned pipelines case, two PEC pipelines of 20 cm diameter are considered with their centroid buried at 80 cm depth and 50 cm lateral displacement (i.e., $d_c = -50$ cm and $h_c = 80$ cm). For the vertically positioned pipelines case, 20 cm diameter PEC pipelines with $d_c = -80$ cm and $h_c = 100$ cm are considered. The Tx–Rx configuration and frequency range considered are the same as in Section IV-A and the medium is the Loam medium in Table I. Algorithm II is applied to the simulation data to predict the pipeline's depth and lateral displacement values. The inversion error is defined as

$$\text{IE} = \sum_{p=1}^{N_p} \sqrt{\left(d_p^{\text{Act}} - d_p^{\text{Pred}}\right)^2 + \left(h_p^{\text{Act}} - h_p^{\text{Pred}}\right)^2} \tag{9}$$

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TABLE II
BENCHMARK PERFORMANCE CASES

Parameter	Case Number																			
	1	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*	*13*	*14*	*15*	*16*	*17*	*18*	*19*	*20*
ar	1	2	2	1	1	1	1	2	2	2	1	1	1	2	2	2	1	2	2	2
N_p	3	5	2	5	5	5	3	4	5	4	3	3	3	3	2	4	4	2	5	2
h_c [cm]	77	87	91	70	71	74	94	79	74	81	73	59	81	89	66	54	79	57	51	70
d_c [cm]	67	34	55	94	52	32	40	2	81	97	74	41	86	27	13	70	96	28	14	99
s [cm]	20	15	20	15	15	25	25	10	15	25	10	25	25	25	25	10	10	15	15	20

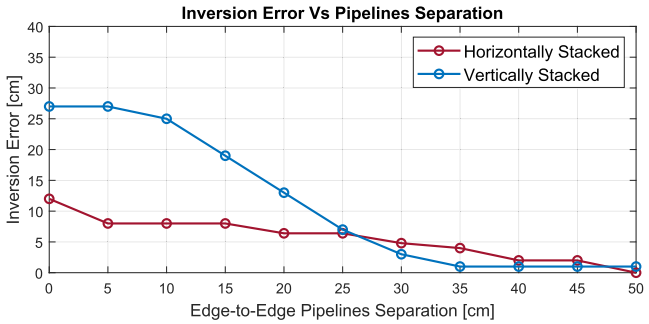


Fig. 11. Inversion error defined in (9) versus varying pipelines separation values. Both horizontally and vertically positioned pipelines are 20 cm diameter PEC pipelines. The pipelines centroid for the horizontally positioned case is ($d_c = -50$ cm and $h_c = 80$ cm) while for the vertically positioned is ($d_c = -80$ cm and $h_c = 100$ cm).

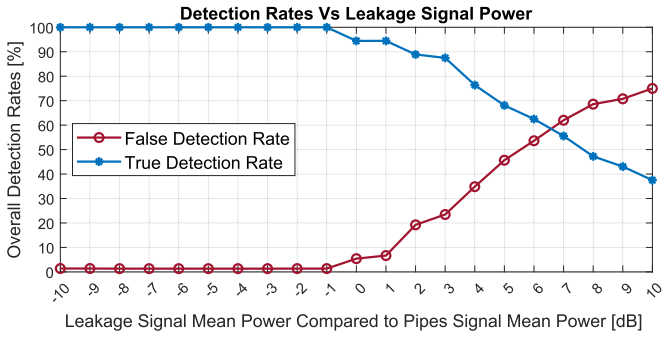


Fig. 12. Overall detection rates across the 20 pipelines cases in Table II buried in sand medium (Table I) for varying levels of leakage signal.

where Act and Pred stand for actual and predicted values by Algorithm II, respectively, and N_p is the number of pipelines. The inversion error is calculated for every separation value for the two cases and the results are shown in Fig. 11. It can be seen that even for small values of edge-to-edge separation, the algorithm maintains an error of several centimeters which is quite good given the operation frequency of the radar (i.e., 150–450 MHz). The error approaches zero as the separation is increased confirming that negligence of multiple interactions in lossy media can still lead to valid inversion due to the media attenuation of the second-order signals.

C. Performance Including Tx–Rx Leakage Signal

The analysis done so far considers the scattered field from the pipelines only. However, in real scenarios for multistatic subsurface radar, the received signal will contain both the response from the buried pipelines as well as a strong leakage signal resulting from the direct coupling between the Tx and Rx in addition to the ground reflection. Such leakage signals could hinder detection if not mitigated. To test the algorithm performance in the presence of leakage signal, the FM in (3) is used to generate pipeline scattered field data. After that, the leakage signal is obtained by full-wave simulation of Tx and Rx Hertzian dipoles located 5 cm above the empty sand medium and added to the FM data. The Tx–Rx configuration is the same one used in Section III-B and the frequency range is from 150 to 450 MHz with a 20-MHz frequency step. Mathematically, the received signal for the mth frequency and nth Rx location is calculated as

$$V_{mn} = \mathrm{FM}_{mn}\left(\mathrm{ar}, N_p, d_c, h_c, s\right) + a\, V_{mn}^{\mathrm{Leakage}} \tag{10}$$

where a is a scaling factor that adjusts the leakage signal to different levels. For a comprehensive analysis, 20 different cases of buried pipelines are considered where the parameters of the FM are obtained randomly and summarized in Table II. To quantify the algorithm performance across all the cases, the true and false detection rates are defined as

$$\text{True Detection Rate} = \frac{N_{\text{Truly Detected}}}{N_{\text{Actual Pipes}}} \tag{11}$$

$$\text{False Detection Rate} = \frac{N_{\text{Falsely Detected}}}{N_{\text{Detected Pipes}}} \tag{12}$$

where the true detection criterion is when the detected pipe location does not differ from the actual location by more than 5 cm and N stands for the total number across all the considered cases. Fig. 12 shows the detection rates for varying levels of leakage signal. The performance results show that the algorithm works well as long as the leakage signal is at the same level as the pipes signal or lower. It should be noted that the leakage signal is typically much stronger than the buried pipeline signals. However, because the leakage signal travels through the air while pipeline signals travel through the soil at lower speeds, the leakage signal arrives much earlier and can be gated out. When proper gating is applied, the remaining part of the leakage signal (the part that has similar delays to the pipeline signals) is only the time domain side-lobes of the leakage signal and such side-lobes typically have low values and are not expected to be higher than the pipeline signals values. When the side-lobes are quite high, a Gaussian window can be applied in the frequency domain to reduce time-domain side-lobes as discussed in [35]. Furthermore, for cases where the pipelines are not buried deep enough, time-gating may not work and alternative differential radar mode can be adopted to mitigate the leakage signal as discussed in [36] or some mechanism to increase isolation between the Tx and Rx may

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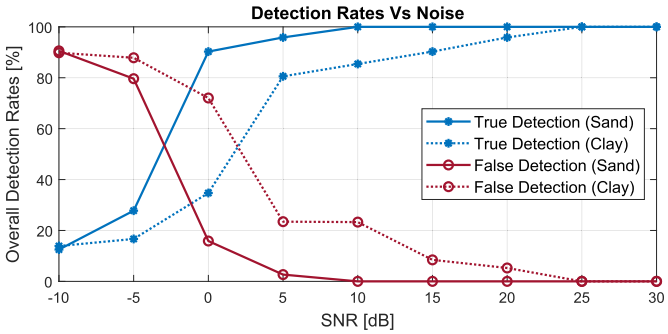


Fig. 13. Overall detection rates across the 20 pipelines cases in Table II buried in sand and clay media (Table I) for varying levels of noise.

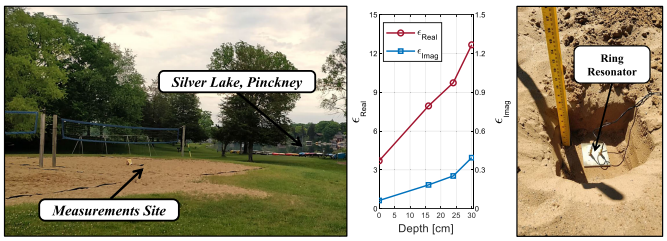


Fig. 14. Experiments location near Silver Lake, Pinckney, MI, USA, along with the ring resonator probe and the resulting dielectric profile.

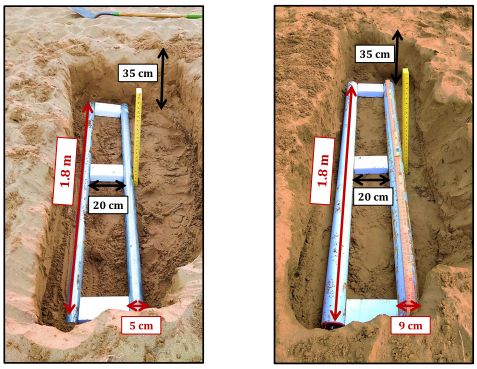
be used as is done in the experimental part of this article (Section IV-E).

D. Performance Evaluation Including Noise

Another aspect of the investigation is the robustness of the dynamic grid search inversion algorithm (Algorithm II) against noise. To investigate that, the data are generated using the FM (3) for the 20 cases in Table II. After that, an additive white Gaussian noise (AWGN) is added to the data with varying signal-to-noise ratios (SNRs). The dynamic grid search algorithm (Algorithm II) in Fig. 5 is then applied to the noisy data—using the same frequency range and Tx–Rx configuration in Section IV-C—and the detection rates defined in (11 and 12) are calculated and plotted in Fig. 13 for the sand and clay media of Table I. The results show the algorithm to be robust as long as the SNR is above 15 dB. It should be noted that such an SNR level can be achieved in subsurface radars operating in the frequency range 150–450 MHz as the propagation losses are not high at these frequencies making the received signal well above the thermal noise level. When the source of noise is radar clutter rather than thermal noise, the algorithm may not perform optimally and some noise suppression procedure needs to be performed before applying the algorithm.

E. Performance Evaluation Using Experimental Data

To evaluate the dynamic grid search algorithm performance on actual data, a realistic sandy environment at Silver Lake Recreation Area, Pinckney, MI, USA, was chosen to perform experiments. The experimental location is shown in Fig. 14 along with the dielectric profile of the sand as a function of depth measured on site by a portable ring resonator probe



(a) (b)

Fig. 15. Buried (a) metallic and (b) water-filled PVC pipelines used in the two experiments.

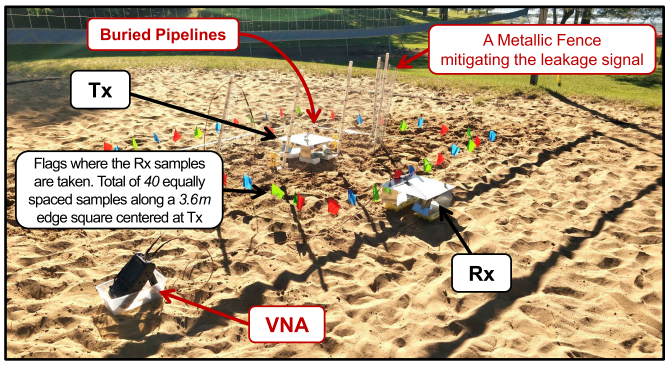
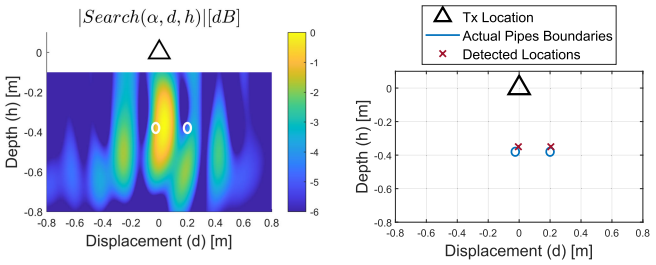


Fig. 16. VNA-based multistatic SAR setup for both experiments. The Rx is moved along a square of 3.6 m edge length collecting S_{21} data at 40 different locations while the Tx is fixed in the middle.

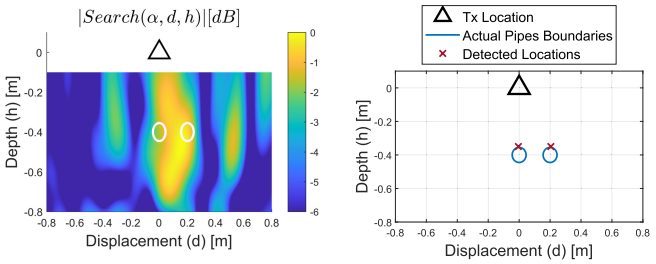
discussed in [37]. After determining the sand permittivity, two experiments were carried out. In one experiment, two metallic pipelines with 5 cm diameter and 20 cm edge-to-edge spacing are buried around 35 cm depth. In the other experiment, two water-filled PVC pipelines with 9 cm diameter and 20 cm edge-to-edge spacing are buried around 35 cm depth. Fig. 15 shows the buried pipelines. The data in the two experiments were collected with a portable vector network analyzer (VNA)-based multistatic SAR system shown in Fig. 16. There is a fixed Tx antenna in the middle of the scene and an Rx antenna that is moved around the Tx along a square of 3.6 m edge length collecting a total of 40 S_{21} parameters samples in different locations through the VNA over the range 150–450 MHz with 20-MHz frequency increments, 3-dBm power (maximum allowed by the VNA), and 1-kHz IF bandwidth. The antennas used for the Tx and Rx are ENVELOPE antenna arrays designed specifically for subsurface radars and discussed in [38]. To mitigate the leakage signal, a metallic fence is inserted between the Tx and Rx antennas. Moreover, the collected S_{21} data are processed to remove the delay and attenuation caused by the coaxial cables. This is done by dividing the collected S_{21} data by S_{21} measured for the cables when they are connected together by a through connector. After removing the cable delay and attenuation, both Algorithm I (Fig. 4) and Algorithm II (Fig. 5) are applied to the data. The results for the metallic pipelines

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(a) (b)

Fig. 17. (a) h-d cut (at the actual α) of the Search space of Algorithm I (Fig. 4) and (b) detection results of Algorithm II (Fig. 5) when applied to metallic pipelines experiment [Fig. 15(a)]. The data are for two metallic pipelines of 5 cm diameter and $d_1 = -22.5$ cm, $d_2 = 2.5$ cm, $\alpha = 0$, and $h_c = -35$ cm (see Fig. 2).



(a) (b)

Fig. 18. (a) h-d cut (at the actual α) of the search space of Algorithm I (Fig. 4) and (b) detection results of Algorithm II (Fig. 5) when applied to water-filled PVC pipelines experiment [Fig. 15(b)]. The data is for two water-filled PVC pipelines of 9 cm diameter and $d_1 = -24.5$ cm, $d_2 = 4.5$ cm, $\alpha = 0$, and $h_c = -35$ cm (see Fig. 2).

TABLE III

COMPUTATION TIME FOR ALGORITHM II AT DIFFERENT SEARCH RESOLUTIONS

$max(N_p)$	$\delta s\,[cm]$	$\delta d\,[cm]$	$\delta h\,[cm]$	Time [minutes]
5	8	8	8	0.97
5	4	8	8	1.52
5	2	8	8	2.8
7	8	8	8	2.03

experiment are shown in Fig. 17. The basic search algorithm results in a single peak between the two pipelines unable to resolve them while the final result of the dynamic grid search algorithm shows two detected pipelines near the actual pipelines boundaries. Moving to the second experiment, the results for the water-filled PVC pipelines are shown in Fig. 18. Here again the dynamic grid search algorithm managed to resolve the two closely spaced pipelines while the basic search algorithm results in a single peak between the two pipelines. A final point to mention is the computation time required by the novel Algorithm II (Fig. 5). The time mainly depends on the required search resolution (i.e., the spacing $\delta s, \delta d$, and δh) and the maximum number of pipelines considered. Using "Intel[1] Core[2] i7-8700 CPU @ 3.20 GHz" processor, 16 frequency points, 36 Rx samples, and fixing $max(s)$ to 80 cm, the required times for running Algorithm II (Fig. 5) at different resolutions are summarized in Table III.

[1]Registered trademark.
[2]Trademarked.

V. CONCLUSION

A novel 3-D forward scattering model and dynamic grid search-based inversion algorithm for multiple buried pipelines in wideband multistatic subsurface SAR is proposed. The model tracks the SPCs of each pipeline for each receiver position and then accounts for both propagation phase and amplitude attenuation due to propagation and medium losses. An objective function is defined as the misfit between the received data and the FM. After that, the dynamic grid search algorithm minimizes the objective function leading to the detection and localization of closely spaced pipelines along the range and cross-range directions. The FM is validated with full-wave simulations and the algorithm is applied on full-wave simulation data as well as experimental data demonstrating its superiority over basic phase compensation (focusing)-based algorithm for buried pipeline detection.

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